Project Initialization and Planning Phase

Date	03 July 2026
Team ID	XXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Modern agriculture depends on several environmental and soil-related factors, making it difficult for farmers to choose the right crop and manage resources efficiently. Many farmers still rely on traditional methods or personal experience, which may not always produce the best results under changing climatic conditions. In addition, researchers and policymakers require reliable agricultural data to understand crop performance and develop sustainable farming strategies.

The OptiCrop project addresses these challenges by providing a smart decision support system that analyzes soil nutrients like Nitrogen, Phosphorus, Potassium, temperature, humidity, pH and rainfall. crop information to generate meaningful recommendations.

By understanding the needs of different users, the project aims to improve agricultural productivity, optimize resource utilization, and support informed decision-making for sustainable farming.

Objectives:

- Understand farmers' challenges in crop selection.
- Analyze soil and environmental data.
- Recommend suitable crops using data-driven insights.
- Improve agricultural productivity and resource utilization.

Expected Outcome:

The system will provide accurate crop recommendations, improve farming decisions,optimize the use of agricultural resources, and support sustainable farming practices.

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Farmer	Choose the most suitable crop for my land	I don't know which crop matches my soil and weather conditions.	I don't have accurate, data-driven recommendations.	Uncertain about achieving maximum yield and profit.
PS-2	Agricultural User	Check whether my land is suitable for a specific crop	it is difficult to evaluate crop suitability manually.	Soil and climate conditions vary across locations.	Confused about selecting the right crop.
PS-3	An Agricultural Researcher/ Policymaker	Analyze crop environment relationships for better planning.	Choose the most suitable crop for my land	Reliable insights are needed for sustainable agricultural planning.	Challenged in making informed decisions.